

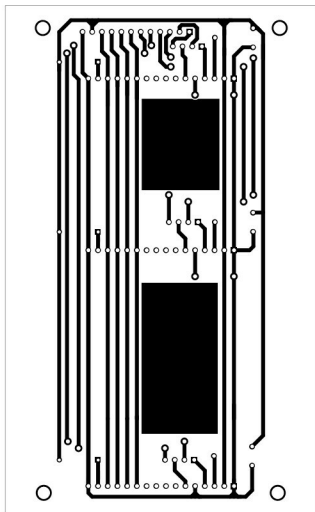


Fig. 2: Actual-size PCB layout for interfacing of multiple LCDs with Arduino

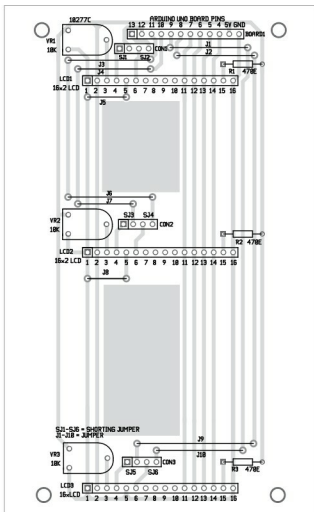


Fig. 3: Components layout for the PCB

LCD3 using SJ5 and SJ6, ensuring that all other RS and EN pins are not used.

When you want to display data on all the three LCDs, connect all the shorting jumpers (SJ1 through SJ6) to the Arduino Uno board. Refer Table II for jumper settings and data displays on different LCDs.

The Arduino board can be powered by an external 9V, 500mA adaptor or USB cable.

Software

The software (multi.ino) for interfacing of the multiple LCDs using Arduino is written in Arduino programming language. The Arduino Uno is programmed using Arduino IDE software.

Construction and testing

An actual-size PCB layout for interfacing multiple LCDs with Arduino is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect Arduino Uno and the PCB using external male-to-male jumpers. Then solder the 16-pin bergstrip male connector on the LCD and 16-pin bergstrip female connector on the PCB. Fix all the LCDs on the

PCB in the space provided.

After assembling the circuit, connect the Arduino board to your computer with standard USB cable. Compile the source code (multi.ino) and upload it to the Arduino Uno board. Connect shorting jumpers SJ1 through SJ6 on the respective connector as explained above. You can see data display on LCD1 or all the LCDs as per the jumper settings (refer Table II). Vary contrast-control preset VR1 left and right until you get clearly visible text on LCD1. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



Jaydeep Tiwari (left) and Shubham Pakhale (right) both are electronics enthusiasts